

Anwar Mousa

Data Engineer

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Summary

Data Engineer with hands-on experience building scalable ETL pipelines, real-time streaming systems, and cloud-based data warehouse solutions — and, increasingly, the AI-powered applications built on top of them. Proficient in Apache Airflow, Kafka, Spark, Snowflake, and Python, with a strong foundation in data modeling, medallion architecture, and LLM-based system design. Proven track record of optimizing pipeline performance and delivering analytics infrastructure that drives business decisions.

Experience

BI Engineer

Jul 2025 – Jan 2026

Link Development · Hybrid

- Designed and delivered end-to-end business intelligence solutions supporting data-driven decision-making across the organization.
- Built and maintained interactive dashboards and executive-level reports for stakeholders using industry-leading BI tools.
- Developed and optimized data models and ETL pipelines to streamline reporting workflows and reduce data latency.
- Collaborated with cross-functional teams to translate complex business requirements into scalable analytical solutions.
- Enforced data quality standards and consistency across all reporting platforms, reducing reporting errors significantly.

Data Engineer

Oct 2024 – May 2025

Digital Egypt Pioneers Initiative (DEPI) · Egypt

- Designed and implemented **ETL pipelines** using **Apache Airflow**, reducing data processing time by **40%**.
- Built real-time streaming data solutions with **Apache Kafka** and **Apache Spark** to handle high-throughput event data.
- Optimized data warehouse queries in **Snowflake**, improving query performance by **35%** through schema redesign and clustering keys.
- Collaborated with cross-functional teams to deliver data-driven insights supporting strategic business decisions.

Data Analyst Intern

Sep 2024 – Oct 2024

NeuronetiX · Hybrid

- Performed statistical analysis on large datasets using **Python** and **SQL** to identify trends and patterns for strategic decision-making.
- Conducted Exploratory Data Analysis (EDA) using **Pandas**, **NumPy**, and **Matplotlib** to extract actionable insights from raw data.
- Created automated reporting dashboards using **Power BI** and **Tableau**, enabling stakeholders to self-serve key metrics.

Education

Bachelor of Computer Science and Informatics

Aug 2022 – Jun 2026

Delta University for Science and Technology · Egypt

- Graduation Project: **Ayn** — AI-native QA & accreditation platform for educational institutions (TypeScript, Next.js, FastAPI, Python) Demo [🔗](#)

Data Analytics Scholarship – Digital Egypt Pioneers Initiative (DEPI)

Oct 2024 – May 2025

Ministry of Communications and Information Technology (MCIT) · Egypt

- Intensive training in data analytics, ETL pipelines, Apache tools, and business intelligence with real-world project delivery.

Projects

Ayn – AI-Native QA & Accreditation Platform

GitHub [🔗](#) — Demo [🔗](#)

- Built a production-grade AI platform integrating **LLMs** as an institutional intelligence layer (Horus) to automate standards compliance, evidence mapping, and gap analysis for educational institutions.
- Developed **FastAPI** backend services and **Next.js** frontend with real-time AI-assisted workflows; deployed on cloud infrastructure with secure multi-tenant architecture.
- Implemented AI-powered analytics and reporting features to surface accreditation insights and reduce manual QA effort.

Real-Time Forex Data Pipeline & Analytics Dashboard

GitHub [🔗](#)

- Built an automated real-time ETL pipeline using **Apache Airflow**, **Apache Kafka**, and **Apache Spark** for processing currency exchange data from Alpha Vantage API.
- Stored and queried processed data in **Snowflake** using a **multi-layer medallion architecture** (Bronze/Silver/Gold) with automated data quality checks.
- Developed an interactive **Streamlit** dashboard featuring candlestick charts, RSI/MACD/EMA indicators, volatility analysis, and custom export options.
- Containerized the full stack with **Docker** for reproducible deployment; implemented secure credential management and structured logging.

Lakehouse Analytics Platform

GitHub [🔗](#)

- Architected a modern lakehouse platform implementing the full **medallion architecture** (Bronze/Silver/Gold) for scalable data processing and ML-ready feature stores.
- Leveraged **Apache Spark** and **Delta Lake** for distributed processing and ACID-compliant storage with schema enforcement.
- Containerized using **Docker** for consistent environment management and easy horizontal scaling.

RideTrack – Real-Time Ride-Sharing Analytics

GitHub [🔗](#)

- Built a real-time ride-sharing analytics platform processing high-throughput trip data streams using **Apache Kafka** and **Apache Spark**.
- Persisted processed events to **PostgreSQL** and visualized KPIs through a **Streamlit** dashboard for operational monitoring.
- Containerized all services with **Docker** following modern data engineering practices for production-grade deployment.

NASA Log Analytics Pipeline

GitHub [🔗](#)

- Processed **3.4M+ log entries (61+ GB)** from NASA HTTP server logs using an end-to-end pipeline built on **Apache Airflow**, **Spark**, and **PostgreSQL**.
- Applied anomaly detection models to identify traffic pattern deviations and error rate spikes at scale.
- Designed interactive monitoring dashboards in **Grafana** to visualize key operational metrics and anomaly detection insights.

Technical Skills

Languages:	Python (Pandas, NumPy, PySpark, Scikit-learn), SQL
Data Engineering:	Apache Airflow, Apache Kafka, Apache Spark, DBT, ETL Pipeline Design
AI & Machine Learning:	Classification Models , Text Analysis, FastAPI, LLM Orchestration, RAG Systems
Cloud & Warehousing:	Snowflake, AWS (EC2, S3, RDS), Delta Lake, Data Lakehouse
Databases:	PostgreSQL, MySQL, Microsoft SQL Server
Data Architecture:	Medallion Architecture, Data Modeling, Dimensional Modeling
Visualization:	Streamlit, Grafana, Power BI, Tableau
DevOps & Tools:	Docker, Git, GitHub, Linux, Bash